

EDUCATION	The Chinese University of Hong Kong <i>Ph.D. in Computer Science and Engineering</i> • Advisor: Prof. Hong Xu • Research area: Machine Learning Systems	Hong Kong SAR, China 2022.08 - 2026.07 (<i>expected</i>)
	Northwestern Polytechnical University <i>B.E. in Computer Science and Technology</i> • GPA: 93.37/100, Rank: 1/247. • National Scholarship twice, First-class Scholarship three times, Outstanding Graduate of Northwestern Polytechnical University.	Xi'an, China 2018.09 - 2022.06
RESEARCH INTEREST	I am broadly interested in systems design for machine learning workloads, especially the following topics: Distributed Machine Learning Systems: Optimizing distributed training for large models, including LLMs, video diffusion models, and RLHF/agent-RL workloads; designing and optimizing inference systems for machine learning workloads. Resource Management and Scheduling: Designing resource management, workload scheduling, and scheduling algorithms for machine learning systems.	
PUBLICATIONS	Preprints - Kaiwen Chen, Xin Tan, Jingzong Li, Hong Xu, Libra: Efficient Resource Management for Agentic RL Post-Training. <i>Preprint</i>. [Preprint] - Yicheng Feng, Xin Tan, Yangtao Deng, Yimin Jiang, Yibo Zhu, Hong Xu, Frontier: Towards Comprehensive and Accurate LLM Inference Simulation. <i>Preprint</i>. [Preprint] [Code] Peer-reviewed Publications [SIGCOMM'26, CCF-A] Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, Dynamic Compute and Network Orchestration for Disaggregated RL. <i>ACM Special Interest Group on Data Communication (SIGCOMM)</i>, 2026. [Preprint] [ASPLOS'26, CCF-A] Xin Tan, Yuetao Chen, Yimin Jiang, Xing Chen, Kun Yan, Nan Duan, Yibo Zhu, Daxin Jiang, Hong Xu, DSV: Exploiting Dynamic Sparsity to Accelerate Large-Scale Video DiT Training. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i>, 2026. [Publication] [Preprint] [Code] [ASPLOS'25, CCF-A] Xin Tan, Yimin Jiang, Yitao Yang, Hong Xu, Towards End-to-End Optimization of LLM-based Applications with Ayo. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i>, 2025. [Publication] [Preprint] [Code] Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, RFabric: A Reconfigurable Network for the Rhythms of Disaggregated RL. <i>The 1st Frontier AI Systems Workshop (FAISys)</i>, 2025. [PDF] [ICPP'24, CCF-B] Xin Tan, Jiamin Li, Yitao Yang, Jingzong Li, Hong Xu, Arlo: Serving Transformer-based Language Models with Dynamic Input Lengths. <i>ACM International Conference on Parallel Processing (ICPP)</i>, 2024. [Publication]	

6. [ICML'26, CCF-A] Kaihua Liang, Xin Tan, An Zhong, Hong Xu, Marco Canini, **FOCUS: DLLMs Know How to Tame Their Compute Bound**. *International Conference on Machine Learning (ICML)*, 2026. [\[Preprint\]](#) [\[Code\]](#)
7. [IWQoS'26, CCF-B] Kaiwen Chen, Xin Tan, Minchen Yu, Jingzong Li, Hong Xu, **MemShare: Memory Efficient Inference for Large Reasoning Models through KV Cache Reuse**. *IEEE/ACM International Symposium on Quality of Service (IWQoS)*, 2026. [\[Preprint\]](#)
8. Yicheng Feng, Xin Tan, Kin Hang Sew, Yimin Jiang, Yibo Zhu, Hong Xu, **Frontier: Simulating the Next Generation of LLM Inference Systems**. *SOSP Workshop on Practical Adoption Challenges of ML for Systems (SOSP-PACMI)*, 2025. [\[Preprint\]](#)

PROJECTS

Compute and Network Co-orchestration for Disaggregated RL

Accepted by SIGCOMM'26; first-author/lead work

2025.05 - 2026.01

- Systematically analyzed disaggregated rollout-training architectures for LLM RL post-training (RLHF/agent RL), identifying generation-stage bottlenecks caused by dynamic output lengths, KV-cache pressure, and long-tail requests, as well as phase-varying communication patterns across training, generation, and weight synchronization.
- Designed OrchestrRL, a compute-network co-orchestration framework with an adaptive rollout scheduler that dynamically switches among TP/EP/AFD parallel configurations based on batch size, generation-length distribution, and remaining request states, and mitigates stragglers through online request migration.
- Designed RFabric, a reconfigurable optical-electrical hybrid fabric that allocates bandwidth on demand for training collectives, generation communication, and weight synchronization; built RLSim, a high-fidelity simulator for evaluating RL clusters across model sizes, parallel strategies, cluster topologies, and workload rhythms.
- Validated OrchestrRL on a 64-GPU NVIDIA H800 cluster, achieving up to 1.42x end-to-end training throughput improvement; large-scale simulation shows RFabric approaches ideal non-blocking Fat-Tree performance and improves performance-cost efficiency by 1.53–2.06x.

High-fidelity Simulation for Modern LLM Inference Serving

Preprint; core contributor

2025.08 - 2026.05

- Analyzed simulation challenges as modern LLM serving evolves from homogeneous monolithic deployments to disaggregated architectures, where existing simulators struggle to jointly model prefill-decode disaggregation (PDD), attention-FFN disaggregation (AFD), complex parallelism, runtime optimizations, and stateful requests.
- Participated in designing and implementing Frontier, a discrete-event simulator that models co-location, PDD, and AFD through role-based cluster workers, incorporates runtime optimizations such as CUDA Graph, speculative decoding, and prefix caching into the scheduler-batch-engine loop, and supports stateful workloads such as reasoning, agents, and RL rollout.
- Contributed high-fidelity cost prediction modules for compute, communication, and memory, covering operator runtime, collective communication, KV-cache capacity, and cross-cluster transfer; Frontier achieved below 4% average throughput error in validation on a 16-GPU NVIDIA H800 cluster and scales to thousand-GPU simulations with SLA-aware design-space exploration.

Accelerating Large-scale Video Generation Training with Attention Sparsity

Published at ASPLOS'26; first-author/lead work

2024.09 - 2025.02

- Analyzed efficiency bottlenecks in large-scale video DiT training, showing that full attention incurs prohibitive cost for high-resolution long-video inputs while dynamic and heterogeneous attention sparsity offers substantial acceleration opportunities.
- Designed and implemented DSV, a training acceleration framework that combines low-rank approximation, two-stage sparsity prediction, and efficient sparse-attention kernels to predict critical key-value pairs and perform sparse computation.
- Extended and optimized context parallelism for sparse attention, including hybrid head/sequence parallelism, load balancing, and communication optimizations; implemented the system with PyTorch FSDP and Triton.
- Evaluated DSV on up to 30B-parameter video DiT models on a 128-GPU cluster, achieving up to 3.02x training throughput improvement while preserving generation quality compared with full attention.

End-to-End Optimization for LLM-based Applications

Published at ASPLOS'25; first-author/lead work

2024.01 - 2024.06

- Analyzed limitations of existing LLM application orchestration frameworks, including limited end-to-end optimization opportunities caused by modular orchestration and request scheduling that cannot directly optimize application-level objectives.
- Designed and implemented Ayo, an end-to-end optimization system centered on fine-grained orchestration, representing workflows across heterogeneous modules as dataflow graphs of task primitives to expose primitive characteristics and interactions for optimizations such as primitive parallelism and pipeline parallelism.
- Implemented Ayo on Ray and evaluated it on LLM applications including search-augmented generation and retrieval-augmented generation, achieving up to 2.09x average end-to-end latency speedup.

Serving Variable-length Language Model Workloads

Published at ICPP'24; first-author/lead work

2022.10 - 2023.06

- Analyzed inefficiencies in serving discriminative language models under variable-length requests, especially the limitations of static compilation with max-length padding and dynamic compilation.
- Proposed Arlo, a hybrid compilation and dynamic scheduling approach that serves requests of different lengths with corresponding statically compiled runtimes and dynamically adjusts GPU allocation according to request-length distributions.
- Designed a multi-level request queueing scheduler to handle short-term distribution shifts and optimize request latency; deployed and evaluated the system with Triton Inference Server and TensorRT, significantly improving efficiency under diverse request distributions.

INTERNSHIPS	System Group, StepFun Beijing, China 2024.09 - 2026.01 - Led research on large-scale video generation training efficiency. Designed and implemented DSV for dynamic attention sparsity in video DiT training, including low-rank approximation and two-stage sparsity prediction, sparse-attention kernels, sparse-aware context parallelism, and load-balancing strategies. The work was published at ASPLOS'26 and achieved up to 3.02x training throughput improvement on a 128-GPU cluster. - Characterized compute-network rhythms and resource bottlenecks across rollout, inference, and training components in large-scale RLHF/agent-RL post-training clusters. Designed dynamic compute-network orchestration and a reconfigurable fabric for RL clusters; built a simulation and evaluation framework supporting different model sizes, parallel strategies, cluster topologies, and workload rhythms. The work was accepted by SIGCOMM'26.
	Network Research Group, Microsoft Research Asia Remote 2021.10 - 2022.03 - Developed a real-time, non-intrusive monitoring system to collect critical AI infrastructure metrics (GPU utilization, network/NVLink bandwidth), and designed a scalable analytics tool to evaluate ML workloads across large-scale clusters. - Analyzed six months of datacenter workload data to characterize resource usage and network patterns of various AI tasks, providing actionable recommendations to optimize cloud infrastructure and software stack.
AWARDS AND HONORS	Student Travel Grant, ASPLOS 2025 2025.04 Full Postgraduate Scholarship, CUHK 2022.08–2026.07 Outstanding Graduate, Northwestern Polytechnical University 2022 First-class Scholarship, Northwestern Polytechnical University 2021 First-class Scholarship, Northwestern Polytechnical University 2020 National Scholarship, Ministry of Education (China) 2020 National Scholarship, Ministry of Education (China) 2019 First-class Scholarship, Northwestern Polytechnical University 2019 Champion, International Underwater Robot Competition, Simulation Track, Team Leader 2020.09 National Third Prize, Loongson Cup National Undergraduate Computer System Design Contest, Team Leader 2020.06
SKILLS	Deep Learning/System Development: PyTorch, vLLM, Megatron, Verl, Triton, TensorRT, CUDA, NCCL Programming: Python, C++, LaTeX English: CET-4 619, CET-6 591, IELTS 7.0
ACADEMIC SERVICES	Reviewer: <i>IEEE Transactions on Network Science and Engineering</i> Shadow Program Committee: <i>ACM EuroSys 2026</i> Artifact Evaluation Committee: <i>USENIX OSDI/ATC 2025, USENIX OSDI/ATC 2024, USENIX FAST 2025, ACM CoNEXT 2025, ACM EuroSys 2025 Spring/Fall</i>
TEACHING	Teaching Assistant: <i>CSCI 3150, Introduction to Operating Systems, CUHK. 2023 Spring, CSCI 1120, Introduction to Computing Using C++, CUHK. 2022 Fall.</i>